MRI-Based Multiple Brain Disease Detection Using Deep Learning

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CERTIFICATE

This is to certify that this project report titled **"MRI-Based Multiple Brain Disease Detection Using Deep Learning"** is a bonafide record of work done by **RAHUL PUDI (228W1A1251) and PAVAN KUMAR KOLLIPATI (228W1A1228)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

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PROJECT SUMMARY

S.No	Item	Description
1	**Project Title**	**MRI-Based Multiple Brain Disease Detection Using Deep Learning**
2	**Student Names & Numbers**	**Rahul Pudi (228W1A1251)** **Pavan Kumar Kollipati(228W1A1228)**
3	**Name of The Guide**	**K. Madhavi**
4	**Research Group**	**Deep Learning**
5	**Application Area**	**Health care**
6	**Aim of the Project**	**To detect brain tumors, Alzheimer's disease, and multiple sclerosis using deep learning with explainable AI on MRI scan images.**
7	**Project Outcomes**	**The project delivers automated MRI-based detection of major brain diseases, enabling timely identification and supporting clinical decision-making. It provides LIME explanations, medical reasoning, and treatment insights to assist healthcare professionals and improve patients health.**

Student Signatures

1.
2.

Signature of the Guide

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LIST OF EQUATIONS

Equation	Brief Description
Eq. [1]	Compound Scaling Rule
Eq. [2]	Compound Scaling Constraint
Eq. [3]	Depthwise Convolution Operation
Eq. [4]	Pointwise Convolution Operation
Eq. [5]	Squeeze Operation
Eq. [6]	Excitation Operation
Eq. [7]	Feature Recalibration
Eq. [8]	Convolution Operation
Eq. [9]	Rectified Linear Unit
Eq. [10]	Max Pooling Operation
Eq. [11]	Hierarchical Feature Learning
Eq. [12]	Fully Connected Layer Activation
Eq. [13]	Dropout Regularization Equation
Eq. [14]	Softmax Function
Eq. [15]	Categorical Cross-Entropy Loss Function
Eq. [16]	Adam Optimizer Update Rule

ABSTRACT

Multiple brain diseases such as brain tumors, Alzheimer's disease, and multiple sclerosis pose severe risks to neurological health and significantly impact cognitive and functional abilities. Early and accurate diagnosis is crucial, yet traditional MRI-based assessment often depends on manual interpretation, which can be slow, resource-intensive, and prone to human error. To overcome these limitations, this project proposes an automated MRI analysis system that detects major brain diseases using deep learning and Explainable AI.

The system employs the EfficientNet-B3 architecture, recognized for its high accuracy and efficiency in medical imaging tasks. Alongside this, AlexNet was implemented to evaluate performance differences, and EfficientNet-B3 demonstrated superior accuracy, making it the primary model for deployment. Trained on diverse MRI datasets, the model accurately classifies various brain tumors (glioma, meningioma, pituitary), Alzheimer's stages (non-demented, very mild, mild, moderate), and multiple sclerosis lesions. By identifying critical visual patterns in MRI scans, the system supports early diagnosis and reduces errors associated with delayed or inconsistent assessments.

To ensure transparency and trust, the system integrates LIME-based Explainable AI, offering visual explanations, medical reasoning, risk scoring, and treatment insights. A user-friendly web interface provides real-time predictions, interpretability features, and hospital locator support for patient assistance. Overall, the proposed framework enhances diagnostic reliability, reduces clinical workload, and supports healthcare professionals through intelligent, data-driven decision-making tools that improve patient outcomes.

Keywords— Deep Learning, EfficientNet-B3, AlexNet, MRI Classification, Brain Tumor Detection, Alzheimer's Disease Detection, Multiple Sclerosis Detection, Explainable AI (XAI), LIME Visualization, Medical Image Analysis, Neural Networks.

CHAPTER-1

Introduction

The chapter delves into the fundamental aspects of the problem addressed in this study, providing a comprehensive overview of its origin, basic definitions, and background. Beginning with an exploration of the problem's inception, this chapter elucidates the context in which it emerged and the factors that prompted its consideration. Subsequently, it establishes foundational definitions and background information essential for understanding the intricacies of the problem domain. Building upon this groundwork, the chapter proceeds to delineate the specific problem statement, articulating the core objectives and challenges inherent in the research endeavor. Furthermore, it highlights the diverse applications and practical implications associated with addressing the identified problem, showcasing its relevance and potential impact across various domains.

1.1Origion of the Problem:

Neurological disorders such as brain tumors, Alzheimer’s disease, and multiple sclerosis are among the most serious medical conditions affecting the central nervous system, with their prevalence increasing rapidly due to aging populations and extended life expectancy worldwide [11]. These disorders lead to progressive deterioration of cognitive, motor, and functional abilities, significantly impacting patients’ quality of life while imposing long-term physical, emotional, and financial burdens on families and healthcare systems. Early and accurate diagnosis is therefore crucial to slow disease progression, enable timely intervention, and improve patient outcomes.

Magnetic Resonance Imaging (MRI) plays a vital role in the diagnosis of neurological diseases because of its ability to capture high-resolution structural and pathological brain details [12]. However, conventional diagnostic procedures rely heavily on manual interpretation of MRI scans by radiologists and neurologists, which is often time-consuming and subject to inter-observer variability. Subtle abnormalities in early disease stages may be overlooked, leading to delayed or incorrect diagnoses. These challenges are further amplified in regions with limited access to experienced medical specialists, where diagnostic efficiency and consistency remain major concerns [12].

Recent advancements in artificial intelligence and deep learning have demonstrated significant potential to address these limitations by enabling automated and reliable analysis of medical images. Deep learning models, particularly convolutional neural networks, can learn complex and discriminative patterns from large-scale MRI datasets, allowing for accurate detection of neurological abnormalities that may not be easily identifiable through manual assessment [13]. Such AI-assisted systems offer faster analysis, improved consistency, and reduced dependency on expert availability, thereby enhancing clinical decision-making.

This project originates from the growing need to integrate intelligent and automated diagnostic solutions into neurological disease detection. By leveraging deep learning models along with explainable AI techniques on MRI datasets, the proposed system aims to accurately identify brain tumors, Alzheimer's disease stages, and multiple sclerosis lesions. The integration of these technologies is intended to assist healthcare professionals, minimize diagnostic delays, and improve accessibility to advanced diagnostic tools, particularly in underserved and resource-limited healthcare environments [11][13].

1.2 Basic definitions and Background:

Basic definition:

1.Brain Tumor:

A brain tumor is an abnormal and uncontrolled growth of cells within the brain or central spinal canal. Tumors may be benign (non-cancerous) or malignant (cancerous) and can originate within the brain (primary tumors) or spread from other parts of the body (secondary tumors). Common types include glioma, meningioma, and pituitary tumors. These tumors disrupt normal brain functions and cause symptoms such as headaches, seizures, and cognitive impairments.

2.Alzheimer's Disease:

Alzheimer's disease is a progressive neurodegenerative disorder that affects memory, thinking, behavior, and overall cognitive function. It is the most common form of dementia and progresses through stages including very mild, mild, moderate, and severe impairment. The disease is characterized by brain tissue shrinkage and the accumulation of abnormal protein deposits such as amyloid plaques and tau tangles. These structural and

biochemical changes gradually disrupt neural communication, leading to cognitive decline and functional deterioration. Early detection is crucial, as timely intervention can help slow progression and improve the patient's quality of life.

3.Multiple Sclerosis:

Multiple sclerosis is an autoimmune neurological disorder in which the immune system attacks the protective myelin sheath surrounding nerve fibers in the central nervous system. This damage disrupts communication between the brain and body, leading to symptoms such as numbness, vision issues, muscle weakness, and coordination problems. MRI scans often reveal characteristic demyelinating lesions.

4.MRI (Magnetic Resonance Imaging):

MRI is a non-invasive imaging method that uses magnetic fields and radio waves to create detailed images of internal body structures. In neurology, MRI is essential for detecting brain disorders due to its ability to visualize soft tissues with high clarity. It is widely used to identify brain tumors, neurodegenerative changes, and MS lesions, and to monitor disease progression.

5.EfficientNet-B7:

EfficientNet-B3 is a high-performance deep learning architecture designed using compound scaling, which uniformly scales network depth, width, and resolution. It offers superior accuracy while maintaining computational efficiency. Its balanced architecture makes it well-suited for medical image analysis tasks, including MRI-based detection of brain diseases.

6.AlexNet:

AlexNet is a pioneering deep convolutional neural network architecture that achieved breakthrough performance in image classification. It introduced innovations such as ReLU activation, dropout, and parallel GPU training. In medical imaging, AlexNet serves as a strong baseline model for MRI classification tasks, allowing comparison of prediction accuracy, feature learning capability, and computational efficiency against more advanced models.

7.Explainable AI (XAI):

Explainable AI includes methods that make AI decisions interpretable and

transparent. In this project, LIME (Local Interpretable Model-Agnostic Explanations) is used to highlight MRI regions influencing the model's decision.

8.Deep Learning:

Deep learning is a subset of machine learning that uses multi-layered neural networks to automatically learn complex patterns from data. In medical imaging, deep learning enables the extraction of subtle features from MRI scans, allowing accurate and consistent disease detection with minimal manual intervention..

9.Image Classification:

Image classification refers to the process of assigning a predefined label to an image based on its visual characteristics. In medical imaging, deep learning-based classification models analyze MRI scans and categorize them into disease types or stages. This helps clinicians make faster and more accurate diagnoses, enabling appropriate treatment planning.

Background:

Neurological disorders such as brain tumors, Alzheimer's disease, and multiple sclerosis are increasingly common due to rising life expectancy and aging populations, leading to significant cognitive, physical, and neurological impairments that often result in long-term disability. Among these conditions, brain tumors are particularly life-threatening because of abnormal and uncontrolled cell growth within the brain, where early symptoms are subtle and difficult to detect, making timely and accurate diagnosis critical [14]. Although Magnetic Resonance Imaging (MRI) is widely used for brain disorder diagnosis, traditional diagnostic procedures rely heavily on manual interpretation by experienced radiologists, which is time-consuming, subjective, and prone to variability due to the complex structure of the brain.

Recent advancements in artificial intelligence, especially deep learning techniques, have shown strong potential in addressing these limitations by enabling automated and accurate analysis of MRI scans. Deep learning models, such as Convolutional Neural Networks and transfer learning–based architectures, can effectively learn complex spatial features and identify subtle abnormalities that may not be easily visible to the human eye, thereby improving early disease detection and diagnostic reliability [15]. These automated

approaches reduce dependency on expert interpretation and support faster, more consistent clinical decision-making in neurological disorder diagnosis.

1.3Problem Statement:

Neurological disorders such as brain tumors, Alzheimer's disease, and multiple sclerosis have become increasingly common and pose major challenges to the healthcare system. These conditions severely impact cognitive, emotional, and physical functioning, often leading to long-term disability. Early diagnosis plays a critical role in improving treatment outcomes and slowing disease progression. However, conventional diagnostic practices rely heavily on manual MRI interpretation, which can be time-consuming and inconsistent. Human error, workload pressure, and the complexity of early-stage symptoms further affect diagnostic accuracy. As the number of cases continues to rise, the need for improved diagnostic tools becomes more urgent.

Delayed detection of brain tumors can result in severe neurological complications and reduced survival rates. Alzheimer's disease and multiple sclerosis progress gradually, often with subtle early symptoms. By the time abnormalities are identified, irreversible damage may have occurred. Traditional diagnosis may fail to differentiate disease stages accurately. These limitations reduce treatment effectiveness and negatively affect patient outcomes.

Access to skilled radiologists is limited, especially in rural or under-resourced regions. The heavy workload on specialists increases the chances of oversight during MRI evaluation. Manual diagnosis depends heavily on expertise and can vary between evaluators. Such inconsistencies delay timely detection and treatment planning. This highlights the need for scalable and consistent diagnostic systems

To address these challenges, automated artificial-intelligence-based methods offer significant potential. This project introduces a deep learning system using EfficientNet-B3 and AlexNet to classify brain tumors, Alzheimer's stages, and multiple sclerosis from MRI scans. The integration of Explainable AI through LIME enhances transparency and provides visual justification for model predictions. Such a system can improve diagnostic accuracy, reduce human error, and support early detection. It strengthens clinical decision-

making by providing consistent and data-driven insights. Ultimately, this solution aims to make advanced diagnostic support more accessible, timely, and reliable.

1.4Applications

1.Clinical Diagnosis Support:

The system assists radiologists and neurologists by providing fast and accurate detection of brain tumors, Alzheimer's disease, and multiple sclerosis from MRI scans. It reduces human error by acting as a reliable second-opinion tool. This enhances confidence in diagnosis and improves overall clinical decision-making.

2.Early Disease Detection:

The model identifies subtle abnormalities that indicate early stages of neurological disorders, enabling quicker diagnosis. Early intervention improves treatment effectiveness and slows disease progression.

3.Telemedicine and Remote Diagnosis:

The system enables remote MRI screening in rural and under-resourced regions where specialists are limited. Physicians can quickly identify high-risk cases and refer patients for advanced care. This expands diagnostic accessibility and reduces unnecessary travel for patients.

4.Research and Development:

Researchers can use the system to analyze large MRI datasets efficiently for studying disease patterns and progression. It helps uncover structural variations across patient groups. This accelerates innovation in medical imaging and supports the development of better diagnostic strategies.

5.Health Monitoring and Screening Programs:

The system can be used in large-scale screening initiatives for early detection of neurological disorders. Automated MRI analysis helps identify abnormal patterns quickly. This strengthens preventive healthcare and improves early intervention outcomes.

6. Hospital and Diagnostic Center Integration:

The system integrates smoothly into hospital workflows, automating MRI image analysis. It reduces radiologist workload and speeds up diagnostic reporting. This improves operational efficiency and enhances overall patient management.

CHAPTER-2

Review of Literature

This chapter conducts a thorough literature survey to explore existing research relevant to the subject matter. By reviewing scholarly works from various sources, including academic journals and conference proceedings, it aims to gather insights and perspectives on the identified problem. Through critical analysis and synthesis of key findings, the chapter seeks to identify trends, gaps, and areas for further investigation. Finally, it provides a concise summary of the literature survey, highlighting key findings and implications for the research.

2.1Description of Existing Systems:

TLEABLCNN is a lightweight convolutional model proposed for detecting brain tumors and Alzheimer's disease using EfficientNet and Squeeze Attention Blocks [1]. The system tackles class imbalance in MRI datasets by applying SMOTE, improving robustness and reducing overfitting. Its architecture includes L2 regularization, batch normalization, global pooling, and dropout to enhance efficiency and stability. Grad-CAM is used to provide explainable visualizations of disease-affected regions, supporting clinical decision-making. The model achieves a high validation accuracy of 95.19%, demonstrating strong generalization. However, the study notes weaker performance in categories like control and non-tumor cases, suggesting areas for future improvement.

The study in [2] presents a system for classifying and detecting Multiple Sclerosis (MS) using MRI images, addressing the challenge of limited and imbalanced medical data. A dataset of 5242 images was balanced through augmentation to improve model reliability. Multiple approaches—including CNN, ResNet-50, CNN combined with ResNet-50, and PCA with SVM—were evaluated for MS detection. Among them, the CNN model achieved the highest accuracy of 99%, as shown in Table 5, outperforming all other models. Confusion matrices in the paper confirm CNN's superior classification performance with minimal errors.

In [4], L. Ramesh, P.S. Pooja Shree, S. Raasika, and B. Rithikashree propose a neural network–based approach for diagnosing Alzheimer's disease using brain MRI

images. Their system applies preprocessing, feature extraction, and a CNN model to classify MRI scans into normal, benign, and malignant categories. According to Tables 1 and 2, the CNN achieves the highest accuracy of 80% and the lowest error rate of 0.4, outperforming SVM and Random Forest. The framework processes MRI images through segmentation and feature engineering before classification. Overall, the study demonstrates that deep learning significantly enhances early Alzheimer's detection and diagnostic reliability.

The study in [3] presents an MRI-based system for detecting Normal, Alzheimer's disease, and Brain Tumor using a lightweight CNN model.A combined dataset from Kaggle was used and divided into training, validation, and testing sets.The proposed 12-layer CNN contains only about 2 million parameters, reducing computational complexity.Its performance was compared with AlexNet, a deeper and heavier model.Results show that the proposed model achieved a maximum accuracy of 98.96%, outperforming AlexNet's 96.90%.Confusion matrix analysis confirms accurate classification with minimal errors, proving the model's effectiveness

The work in [5] by Poornaiah Billa, Bhargavi Nali, Achyuth Karasu, and Ramya Boddapati presents a deep learning–based system for diagnosing multiple brain diseases, including Alzheimer's, brain tumor, epilepsy, and Parkinson's disease. The study utilizes 3D-CNN and deep LSTM models to analyze MRI images and improve diagnostic accuracy and efficiency. Overall, the framework demonstrates that advanced deep learning architectures significantly enhance early detection and classification of neurological disorders.

The study in [6] proposes an improved tumor detection and classification method using a Gaussian Mixture Model (GMM) for precise MRI segmentation and an R-CNN architecture for accurate tumor localization. This combined approach enhances the reliability of tumor detection by improving region extraction and classification performance. Overall, the system demonstrates strong potential for efficient and accurate MRI-based tumor diagnosis.

The work in [8] by Ajitesh Singh, Aarthy S.T., Subham Saha, and Alok Kumar presents a deep-learning system for Multiple Sclerosis detection using U-Net for lesion segmentation and EfficientNet for MRI-based classification. Their model achieves high

segmentation accuracy and 98% classification performance, offering a scalable solution for automated MS diagnosis. This integrated approach enhances clinical decision-making by improving early detection and reducing manual diagnostic burden.

The study in [9] by Vani Lavanya A., Santosh N., and Sai Shree Ram Balaji B. L. introduces an automated brain disease diagnosis and patient support system that uses a CNN model to detect five major brain disorders from MRI scans. Their web-based framework integrates MRI analysis, automated report generation, patient access, and an AI assistant, offering a complete diagnostic workflow. This system enhances diagnostic accuracy and streamlines patient support, contributing to improved healthcare outcomes.

The study in [7] proposes an MRI-based brain disease classification system using EfficientNetB0.MRI datasets for brain tumor, Alzheimer's, and Parkinson's disease were collected and preprocessed. The model was fine-tuned and evaluated for both single and combined disease classification. Results show a maximum validation accuracy of 95.8% in multi-class classification. Confusion matrix analysis confirms reliable classification with fewer parameters, making the model efficient for clinical use.

The study in [10] proposes an MRI-based brain disease classification system using EfficientNetB0.MRI datasets for brain tumor, Alzheimer's, and Parkinson's disease were collected and preprocessed. The model was fine-tuned and evaluated for both single and combined disease classification. Results show a maximum validation accuracy of 95.8% in multi-class classification. Confusion matrix analysis confirms reliable classification with fewer parameters, making the model efficient for clinical use.

2.2Summary of Literature Study

The literature on brain disease detection demonstrates a clear transition from traditional machine learning techniques to advanced deep learning–based diagnostic systems [3], [4]. Early studies primarily employed classifiers such as Support Vector Machines, Decision Trees, and Random Forests, which relied on handcrafted feature extraction from MRI scans [4], [6]. Although these approaches established the groundwork for automated diagnosis, they required extensive preprocessing and showed limited generalization across heterogeneous datasets, thereby restricting their clinical scalability [2], [5]

The introduction of Convolutional Neural Networks (CNNs) marked a significant advancement by enabling automatic extraction of complex spatial and structural features directly from raw MRI images [1], [3]. Deep architectures such as VGGNet, ResNet, and AlexNet substantially improved the accuracy of brain tumor and Alzheimer's disease classification tasks [3], [6]. Furthermore, transfer learning techniques allowed pre-trained models to be efficiently adapted to medical imaging problems with limited data, resulting in improved robustness and reduced training time [7], [10]

Among recent deep learning models, EfficientNet has gained considerable attention due to its high accuracy and computational efficiency [1], [7]. Its compound scaling strategy balances depth, width, and resolution, enabling variants such as EfficientNetB0 and EfficientNetB7 to outperform heavier architectures while using fewer parameters [7], [10]. These characteristics make EfficientNet-based systems particularly suitable for real-time clinical applications and resource-constrained healthcare environments [1], [8].

Recent studies also emphasize the importance of explainability and system integration in clinical workflows [1], [9]. Techniques such as Grad-CAM and LIME have been incorporated to visualize disease-affected regions, enhancing transparency and clinician trust in AI predictions [1]. Additionally, several frameworks integrate automated reporting, patient support systems, and web-based diagnostic platforms, reducing radiologist workload and improving accessibility to neurological diagnostic services [9], [5].

Overall, the evolution from manual, feature-dependent methods to intelligent deep learning–driven systems represents a major advancement in neurological disease detection [2], [8]. These AI-powered solutions deliver higher diagnostic accuracy, faster processing, and improved scalability, enabling earlier disease detection and effective clinical decision support in both advanced hospitals and remote healthcare settings [1], [7], [10].

2.3Software Requirements Specifications

1. Development Environment:

Platform: Kaggle Notebooks (Kaggle Kernels)

Environment: Pre-configured Python environment provided by Kaggle

GPU Support: Optional, used for faster model training and processing.

2. Programming Language:

Python 3.x: Used for scripting all stages of the project, including preprocessing, model building, training, and evaluation.

3. Deep Learning and Machine Learning Libraries:

TensorFlow / Keras: Used to build and train the EfficientNet-B3 model , including preprocessing layers, augmentation, and fine-tuning.

Keras Applications**:** Used for loading the EfficientNet-B3 architecture and initializing ImageNet-pretrained weights.

PyTorch / Keras (AlexNet Implementation): Used in a separate Kaggle notebook for constructing and training the AlexNet model for performance comparison.

Scikit-learn: Used for evaluation metrics such as accuracy, precision, recall, F1-score, confusion matrix, and train–test dataset splitting.

4. Data Handling and Visualization Tools:

NumPy: Used for numerical operations on datasets and arrays.

Matplotlib / Seaborn: Employed for visualizing training performance, loss curves, confusion matrices, and other model evaluation metrics.

Pandas: Used for data manipulation, including loading, cleaning, and transforming dataset contents.

5. Image Processing Libraries:

OpenCV / PIL (Python Imaging Library): Used for image resizing, normalization, and augmentation before feeding the images into the model for training.

CHAPTER-3

Proposed Method

This chapter delineates the design methodology adopted for addressing the research problem, outlining the systematic approach employed in developing the proposed solution. It begins by elucidating the design principles, methodologies, and frameworks utilized to conceptualize and structure the solution architecture. Subsequently, the chapter presents a detailed system architecture diagram, providing a visual representation of the components, interactions, and data flow within the proposed system. Furthermore, it offers comprehensive descriptions of the algorithms employed in the solution, elucidating their functionalities, underlying principles, and implementation details. Additionally, the chapter discusses the datasets and tools utilized in the research, highlighting their significance and relevance in facilitating solution development and evaluation. Through these discussions, the chapter aims to provide a holistic understanding of the design methodology and technical underpinnings of the proposed solution, laying the groundwork for its implementation and evaluation in subsequent chapters.

3.1Design Methodology

The design methodology for this project follows a structured, systematic, and iterative workflow aimed at developing an automated deep learning system for detecting brain tumors, Alzheimer's disease, and multiple sclerosis using MRI scans. The primary objective is to reduce diagnostic errors, support early disease detection, and enhance clinical decision-making through reliable AI-based predictions and explainability techniques. The workflow ensures that each stage—from dataset preparation to deployment—is optimized for accuracy, stability, and real-world applicability.

The methodology begins with dataset acquisition and consolidation, where three separate MRI datasets (Brain Tumor, Alzheimer's Disease, and Multiple Sclerosis) were collected from Kaggle. To maintain class balance and ensure uniform training, 1000 images were selected for each class, including glioma, meningioma, pituitary tumor, MS lesions, normal brain images, and Alzheimer's categories such as non-demented, very mild, mild, and moderate. These datasets were manually cleaned to remove duplicates, blurred

images, and mislabeled samples, ensuring high-quality and diverse MRI samples for training both models.

The next phase involves data preprocessing, where all MRI scans were resized to 224×224 pixels to match AlexNet and EfficientNet-B3 input requirements. Normalization was applied to scale pixel intensity values, and data augmentation techniques such as rotation, horizontal flipping, zooming, and brightness adjustments were used to increase dataset variability and reduce overfitting. After preprocessing, the datasets were merged into a unified structure and split into training, validation, and testing sets to ensure fair and consistent evaluation of both models

Model development was conducted using two independent pipelines, one for AlexNet and another for EfficientNet-B3, each trained in separate Kaggle notebooks. AlexNet served as a baseline model due to its simplicity and historical significance in image classification. EfficientNet-B3 was chosen as the primary model because of its compound scaling strategy, reduced parameter count, and superior performance on medical imaging tasks. Transfer learning was applied for EfficientNet-B3 using ImageNet pretrained weights, and custom dense layers were added for multi-class classification. For both models, cross-entropy loss and Adam-based optimizers were used, along with callbacks such as early stopping and learning rate scheduling to improve convergence and stability.

After model training, each model was evaluated using accuracy, F1-score, confusion matrices, and probability distributions for each disease class. EfficientNet-B3 achieved higher performance and was selected as the final deployed model. To improve model interpretability, LIME (Local Interpretable Model-Agnostic Explanations) was incorporated to generate explanation heatmaps highlighting regions influencing predictions. This explainability component ensures transparency, enabling medical professionals to understand the reasoning behind classifications.

Finally, the system was integrated into a web-based deployment framework developed using Flask, HTML, CSS, and JavaScript. The interface includes features such as disease prediction, LIME visualization, medical reasoning, top-5 probability scores, treatment suggestions, and Google Maps links to nearby hospitals in Vijayawada. The iterative design cycle—consisting of dataset refinement, model tuning, testing, and user-

interface enhancements—ensures the final system is accurate, interpretable, and suitable for clinical decision support in real-world environments

3.2System Architecture Diagram

The system architecture illustrated in Figure 3.1 presents a complete and end-to-end workflow designed for detecting Brain Tumor, Alzheimer's Disease, and Multiple Sclerosis using deep learning and Explainable AI. The architecture is divided into sequential functional blocks, beginning from dataset collection to final prediction and interpretability through LIME visualizations.

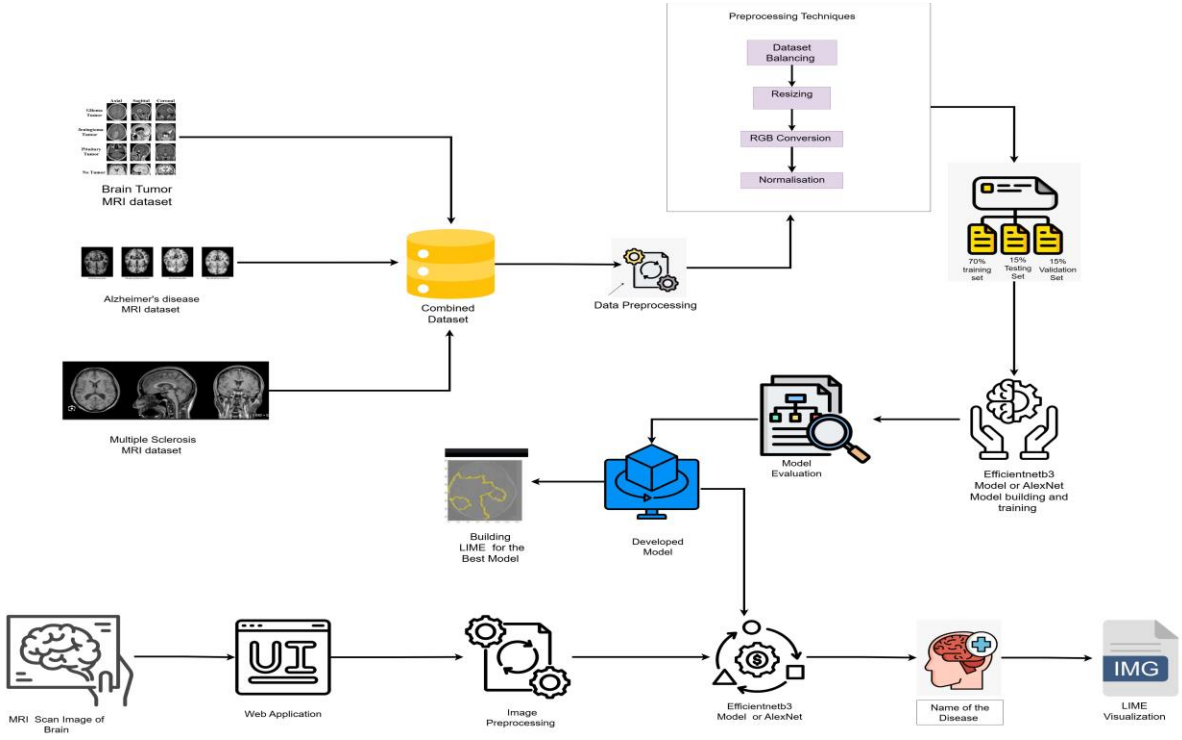

Fig 3.1 System Architecture

The process starts with the acquisition of three separate MRI datasets: the Brain Tumor MRI dataset, the Alzheimer's Disease MRI dataset, and the Multiple Sclerosis MRI dataset. Each dataset contains labeled MRI images belonging to different diagnostic categories such as glioma, meningioma, pituitary tumor, non-demented, mild dementia, MS lesions, and more. These datasets are merged into a combined dataset, ensuring uniform representation by selecting 1000 images per class to maintain class balance.

Once the datasets are combined, they undergo a robust data preprocessing pipeline, which includes dataset balancing, image resizing, RGB conversion, and normalization. These preprocessing techniques ensure consistency in image quality and make the data suitable for model training. After preprocessing, the data is divided into 70% training, 15% testing, and 15% validation, allowing the model to learn effectively and be evaluated fairly on unseen images.

Following preprocessing, the architecture proceeds to model building and training using two different deep learning models—EfficientNet-B3 and AlexNet. Both models are developed in separate training pipelines to compare performance. EfficientNet-B3 is used as the primary model due to its superior accuracy, optimized scaling, and ability to handle complex MRI features. AlexNet serves as the baseline architecture for performance benchmarking. Transfer learning and fine-tuning are applied during training, enabling the models to adapt efficiently to medical imaging characteristics.

Once training is complete, the developed models are passed to the model evaluation stage. Evaluation metrics such as accuracy, precision, recall, loss curves, and confusion matrices are analyzed to determine performance. Based on these results, EfficientNet-B3 is selected as the best-performing model for deployment. The chosen model is then further analyzed using LIME explainability, which highlights the most influential regions in MRI scans to provide transparent and interpretable predictions for medical professionals.

In the deployment phase, a new MRI scan image is uploaded through a web application interface. The input image undergoes preprocessing and is then passed into the trained EfficientNet-B3 model. The model generates predictions showing the identified disease—such as the type of brain tumor, Alzheimer's stage, or presence of Multiple Sclerosis. Alongside the prediction, LIME visualization is generated, showing important regions that contributed to the model's decision, enhancing trust and clinical usability.

Overall, this system architecture ensures a unified, scalable, and automated diagnostic pipeline capable of providing fast, accurate, and explainable predictions. By integrating advanced deep learning models with robust preprocessing and interpretability methods, the system effectively supports medical professionals in the early detection and classification of neurological disorders.

3.3Description of Algorithms

This project employs two deep learning architectures—EfficientNet-B3 and AlexNet—to classify MRI images into Brain Tumor, Alzheimer's Disease, and Multiple Sclerosis categories. EfficientNet-B3 is used as the primary model due to its superior accuracy and efficiency, while AlexNet serves as a baseline for comparison. Additionally, LIME (Local Interpretable Model-Agnostic Explanations) is used to provide interpretability by highlighting the regions that contribute most to the model's predictions.

1. EfficientNet-B3 Algorithm:

EfficientNet-B3 is chosen as the main classifier because of its balanced accuracy and computational efficiency. It is part of the EfficientNet family, which scales model depth, width, and resolution simultaneously using a *compound scaling method*. This allows EfficientNet-B3 to achieve high accuracy without significantly increasing the number of parameters.

EfficientNet employs compound scaling to uniformly scale network depth, width, and input resolution, as expressed in Eq. [1], where a single scaling coefficient controls the overall model size while maintaining optimal accuracy–efficiency trade-offs.

$$\text{Depth} = \alpha^{\phi}, \text{Width} = \beta^{\phi}, \text{Resolution} = \gamma^{\phi} \qquad \text{Eq.[1]}$$

The constraint on compound scaling ensures balanced model growth, as shown in Eq. [2], preventing over-scaling of any single dimension and enabling efficient utilization of computational resources.

$$\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \qquad \text{Eq.[2]}$$

EfficientNet is built using Mobile Inverted Bottleneck Convolution (MBConv) blocks, where depthwise separable convolution significantly reduces computational cost. The depthwise convolution operation is represented in Eq. [3], which applies a single filter per input channel.

$$F_{dw}(i,j,c) = \sum_{m,n} I(i+m, j+n, c) \cdot K_{dw}(m,n,c) \qquad \text{Eq.[3]}$$

The pointwise convolution combines channel-wise features using a 1×1convolution, as shown in Eq. [4], enabling interaction between different feature channels.

$$F_{pw}(i,j) = \sum_c F_{dw}\,(i,j,c) \cdot K_{pw}(c) \qquad \text{Eq.[4]}$$

EfficientNet incorporates the squeeze-and-excitation (SE) mechanism to recalibrate channel-wise feature responses. The squeeze operation computes global channel statistics as given in Eq. [5].

$$s_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i,j,c) \qquad \text{Eq.[5]}$$

The excitation step generates adaptive channel weights using a gating mechanism, shown in Eq. [6], which enhances informative features while suppressing less relevant ones.

$$w_c = \sigma(W_2 \cdot \delta(W_1 \cdot \mathrm{s_c})) \qquad \text{Eq.[6]}$$

The final output of the MBConv block is obtained by rescaling feature maps with the learned channel weights, as described in Eq. [7], improving representational power without increasing complexity.

$$\widehat{F}\,(i,j,c) = w_c \cdot F(i,j,c) \qquad \text{Eq.[7]}$$

These scaled and optimized feature maps allow EfficientNet to achieve high accuracy with fewer parameters, making it highly suitable for medical image classification tasks such as multi-brain disease detection.

The model architecture designed for this project is based on EfficientNet-B3, a high-performance convolutional neural network optimized for medical image classification. The architecture shown in Figure 3.2 illustrates the complete pipeline from input to final classification across 12 disease classes belonging to Brain Tumor, Alzheimer’s Disease, and Multiple Sclerosis categories.

The architecture begins with an Input Layer (244 × 244 × 3), where each MRI scan is resized and converted into a 3-channel RGB format. This ensures compatibility with the pretrained EfficientNet-B3 model. The input image is then passed into the core feature extractor, EfficientNet-B3, which contains 449 layers and is initialized with ImageNet pretrained weights. To adapt the model to MRI image patterns, fine-tuning is applied starting from block6a_expand_conv, enabling deeper layers to learn disease-specific structural features.

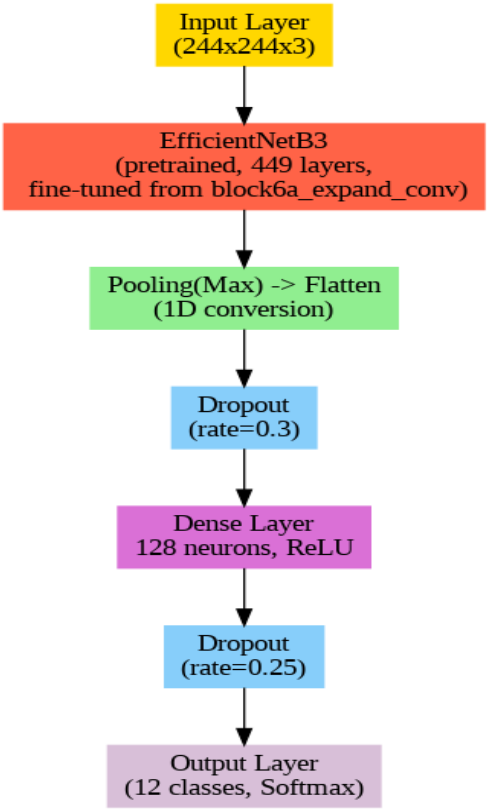

Fig3.2 EfficientNet B3 Model Layers

After feature extraction, the output from EfficientNet-B3 undergoes Max Pooling, followed by a Flatten layer to convert multidimensional feature maps into a 1-dimensional vector suitable for classification. A Dropout layer with a rate of 0.3 is applied to reduce overfitting by randomly disabling a fraction of neurons during training. This is followed by a Dense Layer containing 128 neurons with ReLU activation, which learns high-level representations of MRI features relevant to disease categories. Another Dropout layer (rate = 0.25) is added to further improve generalization and prevent model overfitting.

The final stage of the architecture is the Output Layer, a fully connected softmax classifier with 12 output neurons, corresponding to the 12 combined classes across the three diseases. The softmax activation generates class-wise probability scores, enabling the model to identify the most likely disease category present in the input MRI scan.

2. AlexNet Algorithm:

AlexNet is implemented as a baseline model for comparison. It is a classical CNN architecture known for pioneering deep learning in image classification.

The convolution operation used to generate feature maps is expressed in Eq. [8], enabling the network to detect edges, textures, and disease-specific patterns.

$$F(i,j) = \sum_m \sum_n X(i+m, j+n) \cdot K(m,n) \qquad \text{Eq.[8]}$$

Non-linearity is introduced using the Rectified Linear Unit (ReLU), which improves training efficiency and prevents vanishing gradient problems. The ReLU activation function is defined in Eq. [9].

$$\mathrm{ReLU}(x) = \max(0, x) \qquad \text{Eq.[9]}$$

AlexNet employs max-pooling layers to reduce spatial dimensions while retaining the most prominent features. The max-pooling operation is mathematically represented in Eq. [10], which selects the maximum activation within a local neighborhood.

$$P\ (i,j) = \max_{(m,n)\in\Omega} F(i+m, j+n) \qquad \text{Eq.[10]}$$

As the network depth increases, deeper convolutional layers learn more abstract representations by stacking multiple convolution and activation operations. This hierarchical feature learning is expressed in Eq. [11].

$$F^{(l)} = \mathrm{ReLU}(F^{(l-1)} * K^{(l)}) \qquad \text{Eq.[11]}$$

The fully connected layers perform high-level reasoning by combining features learned from all spatial locations. Neuron activation in these layers is computed as shown in Eq[12]

$$h = \mathrm{ReLU}(Wz + b) \qquad \text{Eq.[12]}$$

To reduce overfitting caused by the large number of parameters in AlexNet, dropout regularization is applied by randomly deactivating neurons during training. This regularization technique is expressed in Eq. [13].

$$h' = h \cdot r, r \sim \mathrm{Bernoulli}(p) \qquad \text{Eq.[13]}$$

For multi-class classification, AlexNet employs the Softmax function to convert the final layer activations into probability distributions across classes. The Softmax function is defined in Eq. [14].

$$P\ (y = i) = \frac{e^{z_i}}{\sum_{j=1}^{N} e^{z_j}} \qquad \text{Eq.[14]}$$

Training minimizes the categorical cross-entropy loss shown in Eq. [15], where the objective is to reduce the discrepancy between predicted probabilities and true class labels.

$$\mathcal{L} = -\sum_{i=1}^{N} y_i \log(\hat{y}_i) \qquad \text{Eq.[15]}$$

Parameter updates during training are performed using the Adam optimizer, which adapts learning rates based on first- and second-order moments of gradients. The update rule is given in Eq. [16].

$$\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \qquad \text{Eq.[16]}$$

This formulation highlights the internal working of AlexNet while excluding explicit input and output layer representations, making it suitable for concise academic reporting.

Key Characteristics:

- 5 convolutional layers
- 3 fully connected layers
- Uses ReLU and MaxPooling
- Supports dropout to reduce overfitting

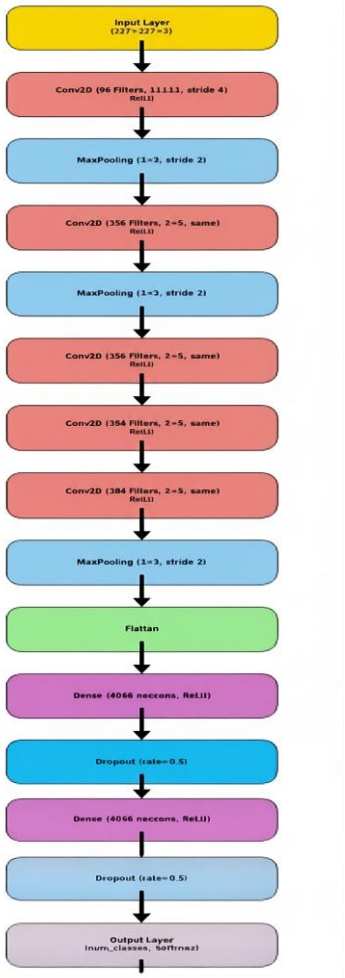

Fig 3.3 AlexNet Model Layers

The AlexNet architecture illustrated in Figure 3.3 represents the baseline deep learning model used in this project to classify MRI images belonging to Brain Tumor, Alzheimer's Disease, and Multiple Sclerosis categories. AlexNet follows a sequential convolutional pipeline designed to extract increasingly complex spatial features from MRI scans before passing them into fully connected layers for final classification.

The architecture begins with an Input Layer (227 × 227 × 3), where MRI images are resized to meet the standard AlexNet input dimensions. The input is then fed into the

first convolutional block, consisting of a Conv2D layer with 96 filters, a kernel size of 11 × 11, and a stride of 4, followed by a ReLU activation function. This layer captures broad visual patterns such as tissue boundaries and structural variations in the brain. A MaxPooling layer (3 × 3, stride 2) is applied to reduce the spatial dimensions while retaining essential features.

The second convolutional stage applies 256 filters with a kernel size of 5 × 5 (same padding), enabling the model to detect medium-scale textures and features relevant to neurological abnormalities. Another MaxPooling layer further downsamples the feature maps. This is followed by three successive convolutional layers with 384, 384, and 256 filters, each using a 3 × 3 kernel (same padding). These layers specialize in extracting deep, fine-grained features such as tumor shapes, Alzheimer's cortical shrinkage patterns, and MS lesion textures.

Once feature extraction is complete, the output is passed to a Flatten layer, converting the multidimensional feature maps into a 1D vector. This vector is then processed by two Dense layers, each containing 4096 neurons with ReLU activation, enabling high-level feature learning. To prevent overfitting, Dropout layers with a rate of 0.5 are applied after each Dense layer, ensuring better generalization to unseen MRI data. The architecture ends with a final Output Layer, containing neurons equal to the number of disease classes in the dataset. A softmax activation function generates probability scores for each class, allowing the model to determine the most likely disease category present in the input MRI scan.

Overall, the AlexNet architecture provides a solid baseline for comparison with EfficientNet-B3. While simpler and heavier in parameters, AlexNet remains effective in capturing structural MRI features, enabling meaningful performance benchmarking in this project.

3. LIME Explainability Algorithm:

To enhance transparency and trust, the project integrates LIME (Local Interpretable Model-Agnostic Explanations) for visual interpretation.

How LIME Works:

1. Takes the MRI image and generates perturbed samples

2. Observes how predictions change when certain pixels are masked
3. Builds a local interpretable model (e.g., linear regression)
4. Highlights important superpixels influencing the prediction

3.4Description of datasets and Tools

Description of datasets:

The project utilizes three MRI-based medical imaging datasets to develop deep learning models for the classification of Brain Tumor, Alzheimer's Disease, and Multiple Sclerosis. These datasets are sourced from publicly available Kaggle repositories and were selected for their high quality, diversity of MRI scans, and strong relevance to neurological disease detection. Each dataset contains labeled MRI images that represent distinct diagnostic categories, allowing the system to learn comprehensive patterns across multiple brain disorders.

For brain tumor detection, the Brain Tumors MRI Dataset is used. This dataset contains thousands of MRI images categorized into four major classes: Glioma, Meningioma, Pituitary Tumor, and No Tumor. The dataset offers varying orientations, contrast levels, and tumor shapes, enabling the model to generalize well across different clinical cases. Preprocessing steps applied to this dataset include resizing, normalization, contrast adjustment, and data cleaning to eliminate duplicate or corrupted images. To maintain dataset balance for model training, 1000 images were selected for each tumor class and prepared for further processing.

For Alzheimer's disease classification, the project employs the Augmented Alzheimer MRI Dataset, which contains MRI scans categorized into four cognitive stages: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. The dataset includes a wide range of augmented images, enhancing variability and improving the model's ability to recognize subtle structural differences in the brain. Preprocessing techniques such as resizing, skull stripping, noise reduction, and normalization are applied to standardize the dataset. A balanced subset of 1000 images per Alzheimer's class is used to ensure equal representation during model training.

For the detection of Multiple Sclerosis, the Multiple Sclerosis MRI Dataset is integrated into the combined dataset. This dataset includes MRI scans showing clear MS

lesions as well as scans of healthy brains, allowing the system to distinguish between normal tissue and demyelination patterns characteristic of MS. The dataset contains both T1 and T2-weighted MRI sequences, which provide valuable contrast variations for lesion identification. Preprocessing steps include lesion-focused cleaning, resizing, grayscale normalization, and artifact removal. As with the other datasets, 1000 images are selected for each MS class (MS vs. Normal) to maintain balanced training conditions.

Together, these three datasets form a unified and balanced combined dataset consisting of 12 total classes across all diseases. Their diversity, rich feature variations, and structured labeling enable the development of a high-performance deep learning system capable of detecting multiple neurological disorders from MRI scans. The careful selection of 1000 images per class ensures fairness, reduces class imbalance, and supports accurate and reliable classification performance. Overall, this combined dataset forms the backbone of the project, enabling a scalable, robust, and real-world-ready AI-driven diagnostic solution.

Tools

The implementation of this project relies on a comprehensive set of tools, frameworks, and libraries that collectively enable efficient data processing, model training, and system deployment. Python 3.x serves as the primary programming language because of its flexibility and the extensive ecosystem of machine learning and image-processing libraries available. The models—EfficientNet-B3 and AlexNet—were trained in separate Kaggle Notebook environments, each offering a pre-configured Python setup with optional GPU acceleration to handle the computational cost of deep learning on large MRI datasets

For deep learning model development, the project utilizes TensorFlow and Keras, which provide high-level APIs for building, fine-tuning, and training convolutional neural networks. EfficientNet-B3 was implemented using the tf.keras.applications module with transfer learning, leveraging ImageNet pretrained weights for faster convergence and improved accuracy on medical imaging tasks. AlexNet was implemented manually in a separate notebook using Keras layers, serving as a baseline model for comparison. These tools made it possible to design flexible architectures, modify hyperparameters, and monitor training performance efficiently.

The project incorporates scikit-learn for classical machine learning utilities, including dataset splitting, label encoding, and performance evaluation. Metrics such as accuracy, precision, recall, F1-score, and confusion matrices were generated using scikit-learn's evaluation modules. These metrics provide a detailed understanding of each model's strengths, weaknesses, and generalization ability across multiple disease classes.

For data manipulation, cleaning, and merging of the three Kaggle datasets, Pandas and NumPy are used extensively. These libraries facilitate tasks such as combining datasets, selecting 1000 images per class, handling file paths, normalizing pixel arrays, and performing numerical operations required for preprocessing pipelines. Visualization tools such as Matplotlib and Seaborn play a key role in generating accuracy curves, loss graphs, confusion matrices, probability bar charts, and LIME heatmap visualization plots.

Real-time image preprocessing and augmentation during training are handled by ImageDataGenerator from Keras, which performs scaling, normalization, rotation, flipping, zooming, and resizing to match model input dimensions (224×224 for EfficientNet-B3 and 227×227 for AlexNet). These augmentations help reduce overfitting and improve the robustness of the models when dealing with new MRI scans.

For model explainability, the project uses LIME (Local Interpretable Model-Agnostic Explanations) to generate interpretable heatmaps that highlight regions influencing the EfficientNet-B3 model's predictions. This enhances transparency and supports clinical use by allowing radiologists to understand the basis of AI decisions.

Finally, the trained EfficientNet-B3 model is deployed using the Flask web framework, where HTML, CSS, and JavaScript are used to design an intuitive user interface. The interface includes image upload functionality, prediction display, top-5 probability charts, medical reasoning, treatment suggestions, and Google Maps integration for nearby hospitals. Together, these tools form a powerful and scalable pipeline for multi-disease MRI analysis and real-time AI-assisted diagnosis.

Together, these tools create a powerful end-to-end diagnostic pipeline capable of handling large-scale MRI data and delivering accurate, explainable predictions.

CHAPTER-4

Results and Observations

This chapter presents a comprehensive analysis of the results obtained from the implementation and evaluation of the proposed solution. It begins by providing a stepwise description of the results, outlining the procedures followed and the outcomes achieved at each stage of the research process. Through this detailed account, readers gain insights into the methodological rigor and effectiveness of the research approach. Subsequently, the chapter delves into a thorough analysis of the results, examining key findings, trends, and patterns emerging from the data. It elucidates the implications of these results, drawing connections to the research objectives and broader implications for the field. Additionally, the chapter offers observations derived from the work, highlighting noteworthy insights, challenges encountered, and areas for further exploration. Through this synthesis of results and observations, the chapter provides a comprehensive understanding of the research outcomes and their significance in advancing knowledge in the domain.

4.1Stepwise description of Results

Fig4.1 illustrates the training and validation accuracy curves for the EfficientNet-B3 model over 10 epochs. The graph shows a consistent upward trend in both accuracies, indicating that the model is learning meaningful representations from the MRI data during training. The training accuracy increases steadily from approximately 0.57 in the first epoch to around 0.96 by the final epoch, demonstrating strong convergence and effective optimization.

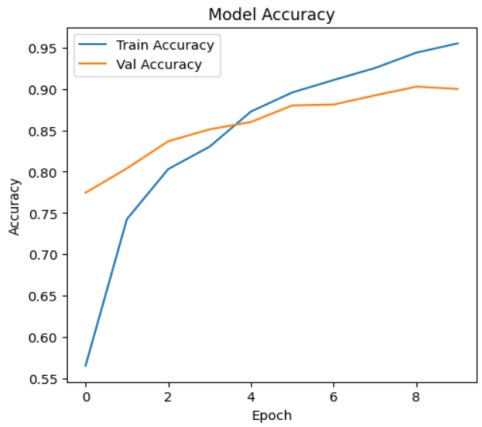

Fig 4.1 EfficientNet B3Accuracy Graph

The validation accuracy also improves significantly during the early epochs, rising from around 0.78 to approximately 0.90, reflecting the model's ability to generalize well to unseen data. The gap between training and validation accuracy remains moderate, suggesting that the model does not suffer from severe overfitting. Instead, EfficientNet-B3 maintains stable generalization performance, which is crucial for medical image classification tasks.

Overall, the accuracy curve indicates that EfficientNet-B3 achieves high performance and successfully learns discriminative features from brain MRI images across multiple disease classes. This demonstrates the model's suitability for deployment in a real-time diagnostic system.

Fig4.2 shows the training and validation loss curves for the EfficientNet-B3 model across 10 epochs. The training loss exhibits a sharp decline from an initial value of approximately 1.25 to around 0.10, indicating that the model quickly learns meaningful features and continues to optimize effectively over successive epochs. This consistent downward trend reflects strong convergence and efficient feature extraction.

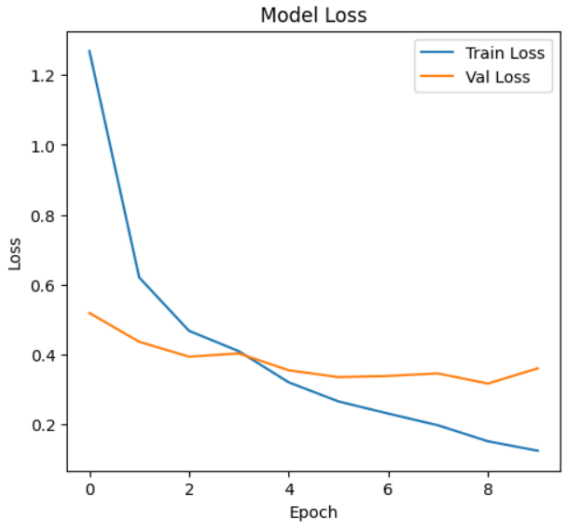

Fig 4.2 EfficientNet B3 Loss Graph

The validation loss also decreases during the early epochs, starting around **0.55** and dropping to approximately 0.35. Although it does not decline as sharply as the training loss, it remains relatively stable with minor fluctuations, suggesting that the model maintains good generalization. The small gap between training and validation loss

demonstrates that the model does not suffer from significant overfitting, even when trained on multiple disease classes.

Overall, the loss curve indicates that EfficientNet-B3 achieves robust learning with a balanced reduction in both training and validation errors. This confirms the model's reliability and suitability for accurately classifying MRI images for brain tumor, Alzheimer's disease, and multiple sclerosis detection.

Fig 4.3 presents the confusion matrix for the EfficientNet-B3 model, illustrating its performance across all 12 disease classes, including Brain Tumor types, Alzheimer's stages, and Multiple Sclerosis categories. The diagonal cells contain high values for most classes, which indicates that the model correctly classifies a large portion of the input MRI images.

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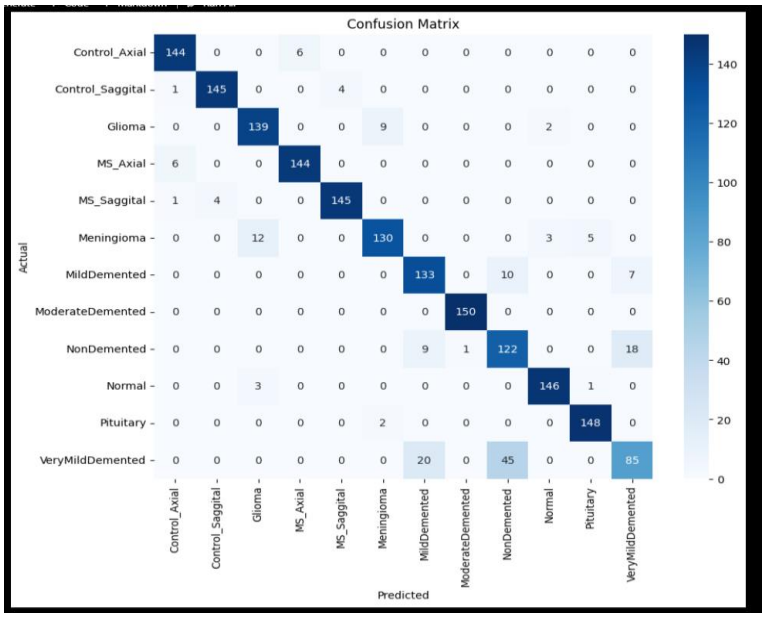

Fig 4.3 EfficientNet B3 Confusion Matrix

The model demonstrates excellent performance in distinguishing MRI orientations and disease categories such as Control_Axial, Control_Saggital, Glioma, MS_Axial, MS_Saggital, Meningioma, Normal, and Pituitary, all of which show classification counts exceeding 130 correct predictions. These strong diagonal patterns reflect EfficientNet-B3's ability to extract fine-grained structural and texture-based features in MRI scans.

For Alzheimer's disease stages, the model performs consistently well for Mild Demented, Moderate Demented, and Very Mild Demented, though slight misclassifications occur among neighboring cognitive stages. This is expected due to subtle differences in structural brain changes across early Alzheimer's progression. For example, VeryMildDemented shows minor confusion with NonDemented and MildDemented, reflecting the overlap in early-stage clinical patterns. The confusion matrix also highlights that misclassifications across completely unrelated disease categories are minimal, demonstrating high discriminative capability. Very few brain tumor images are misclassified into Alzheimer's or MS categories, and vice versa, suggesting strong robustness of the model in multi-disease classification. Overall, the EfficientNet-B3 model displays a high level of accuracy and reliability, achieving strong predictions across all classes with only minimal confusion in structurally similar or clinically overlapping categories. This confirms the model's capability to serve as an effective diagnostic tool for multi-disease brain MRI analysis.

Figure 4.4 illustrates the training and validation accuracy curves for the AlexNet model across 20 epochs. The training accuracy shows a steady upward trend, beginning at approximately 0.30 in the initial epoch and gradually increasing to around 0.82 by the final epoch. This indicates that the model is effectively learning meaningful MRI features and improving its discriminative capability over time.

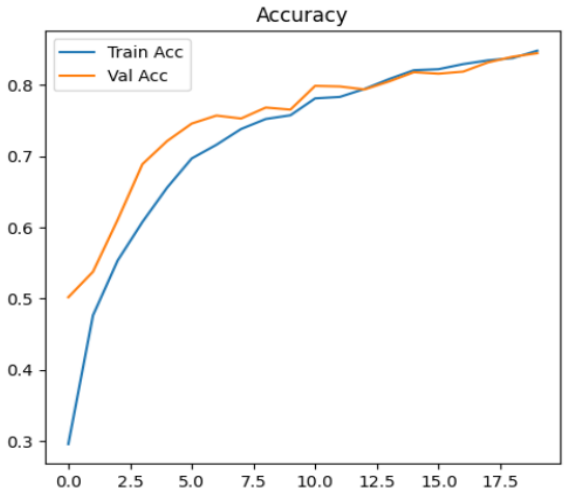

Fig 4.4 AlexNet Accuracy Graph

The validation accuracy follows a similar pattern, starting near 0.50 and rising consistently to reach approximately 0.83, closely aligning with the training accuracy curve. The close alignment between the two curves suggests that the model maintains good generalization and does not suffer from significant overfitting. The early rapid increase in validation accuracy reflects AlexNet's ability to quickly capture essential low-level and mid-level features from the MRI images. Overall, the accuracy graph demonstrates that AlexNet achieves stable and reliable performance across multiple disease classes, although its peak accuracy is lower compared to the EfficientNet-B3 model. Nonetheless, it serves as a strong baseline model and validates the effectiveness of deep convolutional architectures for multi-disease MRI classification.

Figure 4.5 presents the training and validation loss curves for the AlexNet model across 20 epochs. The training loss decreases sharply from approximately 1.75 in the first epoch to around 0.38 by the final epoch, indicating that the model successfully learns meaningful hierarchical features from the MRI data. This continuous downward trend reflects effective optimization and a stable learning process.

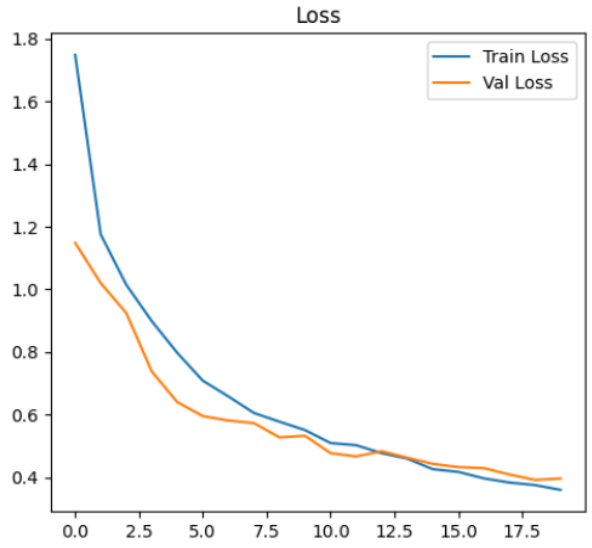

Fig 4.5 AlexNet Loss Graph

The validation loss also shows a consistent decline, starting at around 1.20 and decreasing to approximately 0.36 over the training epochs. The close proximity between the training and validation loss curves suggests that AlexNet generalizes well to unseen

data without significant overfitting. Minor fluctuations in the validation curve are expected in multi-class medical image datasets, yet the overall trend remains smooth and stable.

Overall, the loss graph demonstrates that AlexNet achieves robust convergence and maintains a balanced reduction in both training and validation errors. This confirms that the model is capable of learning discriminative MRI features effectively, though its performance remains lower than the more advanced EfficientNet-B3 model used in this project.

The confusion matrix in Figure 4.6 shows that AlexNet performs reasonably well across several classes, particularly Control_Axial, Control_Saggital, MS_Axial, MS_Saggital, Meningioma, ModerateDemented, Normal, and Pituitary, where most predictions fall along the diagonal. This indicates that the model can correctly identify categories with stronger structural differences in MRI scans.

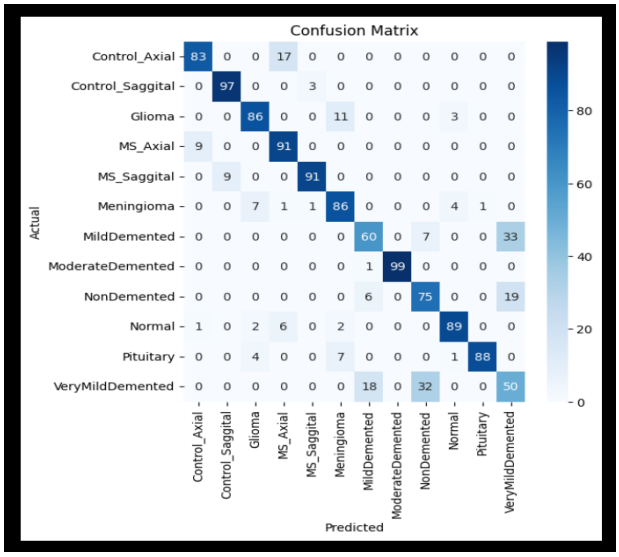

Fig 4.6 AlexNet Confusion Matrix

However, the model struggles with Alzheimer's-related classes such as NonDemented, VeryMildDemented, and MildDemented, where subtle structural variations lead to noticeable misclassifications. Early-stage Alzheimer's features are harder for AlexNet's shallow architecture to capture, resulting in overlaps between adjacent cognitive stages.

Overall, the matrix shows that AlexNet performs well on clearly distinguishable MRI categories but has limited ability to differentiate subtle patterns, reaffirming that deeper models like EfficientNet-B3 provide more reliable multi-disease classification.

Web Application:

The web interface includes a simple and user-friendly MRI Image Upload Module that allows users to upload brain MRI or medical images for analysis. The interface provides a file selection button and an Analyze button, guiding users through the workflow with clear instructions. Once an image is uploaded, the system processes it and generates predictions, LIME visualizations, risk scores, and recommended treatments.

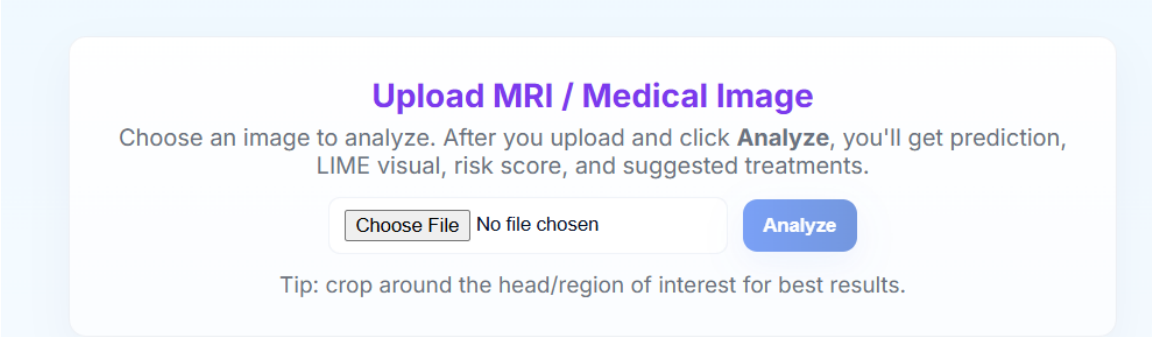

Fig 4.7 MRI Image Upload Interface

The figure 4.8 illustrates the original MRI image alongside the LIME-generated visual explanation, which highlights regions that influenced the model's prediction. The yellow-marked areas indicate abnormal textures or structural deviations associated with the detected condition. This interpretability feature helps clinicians understand why the model arrived at a specific diagnosis.

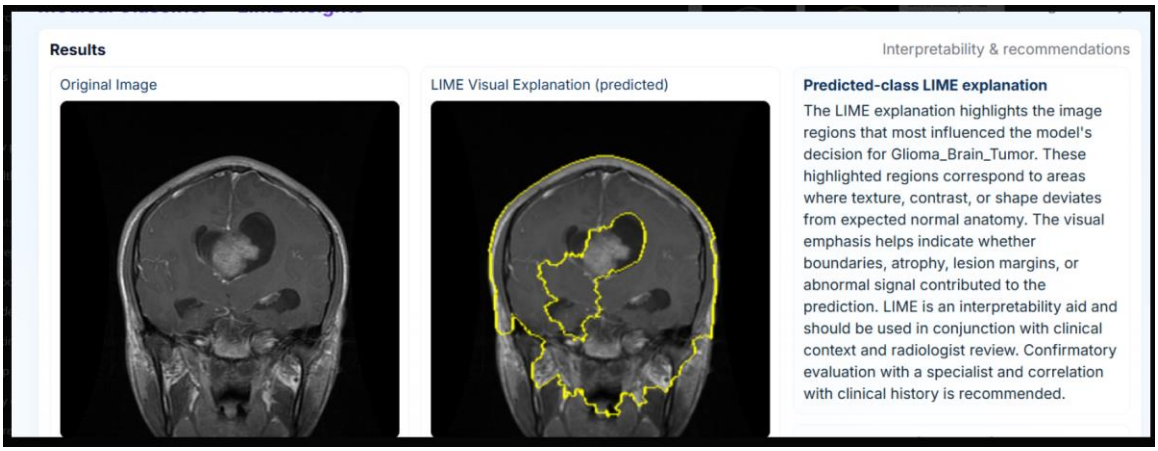

Fig 4.8 LIME-Based Visual Explanation of MRI Prediction

The figure 4.9 presents the calculated risk score, indicating the model's confidence and estimated severity for the predicted condition. It also displays the top five predicted classes, helping users understand alternative possibilities in the diagnosis. Additionally, the

interface provides nearby hospital links in Vijayawada, enabling quick access to medical consultation based on the prediction

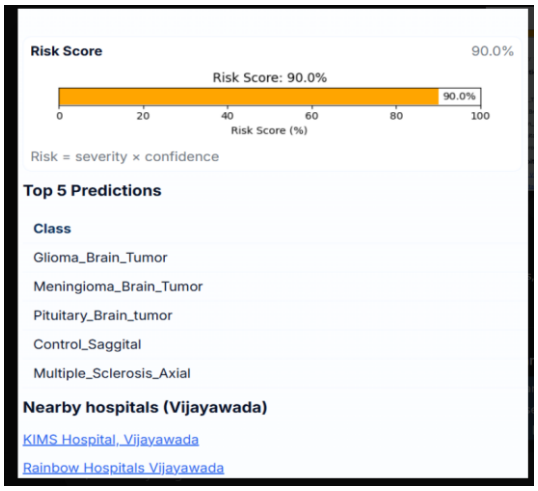

Fig 4.9 Risk Score, Predictions, and Nearby Hospitals

4.2Result Analysis and Test case results

Result Analysis

The performance evaluation of both models reveals a clear distinction in their ability to classify MRI-based brain diseases accurately and consistently. EfficientNet-B3 demonstrates strong superiority over AlexNet, achieving high precision, recall, F1-score, and accuracy across almost all disease categories, including brain tumors, Alzheimer's stages, and Multiple Sclerosis orientations. This indicates that EfficientNet-B3 is far more capable of capturing subtle structural variations present in medical images, such as cortical thinning in Alzheimer's or lesion intensity differences in tumor cases.

For critical clinical classes such as Moderate Demented, Pituitary Tumor, Normal Brain, and MS Sagittal/Axial orientations, EfficientNet-B3 consistently achieves accuracy levels between 96% and 100%, reflecting excellent feature extraction and generalization. In contrast, AlexNet exhibits limitations in learning fine-grained patterns, with noticeably lower accuracy in early-stage Alzheimer's classes such as Very Mild Demented and Mild Demented, where it struggles to exceed 50–60%. These classes require a deeper understanding of subtle anatomical deviations—something EfficientNet-B3 handles more effectively due to compound scaling and deeper representational capacity.

Brain tumor categories such as Glioma, Meningioma, and Pituitary show further evidence of this improvement. EfficientNet-B3 achieves around 93–99% accuracy, whereas AlexNet remains within 86–88%, highlighting its inability to distinguish nuanced tumor margins and textures consistently. Additionally, for Multiple Sclerosis classifications, EfficientNet-B3 maintains accuracy above 96%, confirming its reliability in detecting white-matter lesion variations that AlexNet fails to capture with the same precision.

Overall, the performance patterns indicate that EfficientNet-B3 provides smoother convergence, stronger generalization, and more balanced predictions across all disease types. Its superiority over AlexNet reinforces its suitability for real-world clinical environments where reliability and precision are crucial. The result analysis confirms that the proposed system, powered by EfficientNet-B3, is robust, accurate, and highly effective for assisting in diagnostic workflows involving MRI-based neurological disease detection.

Test Case Results

The fig 4.10 represents test case demonstrates how the system processes a pituitary tumor MRI scan and generates an interpretable explanation using LIME. The original MRI is displayed beside the LIME visualization, where the yellow-outlined regions highlight the areas most influential in the model's prediction.

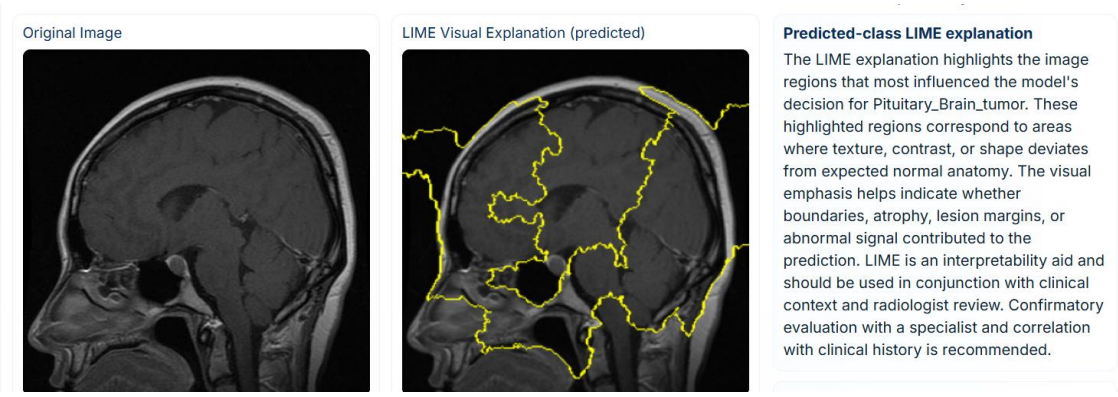

Fig4.10 Output Showing Original MRI, LIME Explanation-1

The explanation panel clarifies why the model identified the case as a Pituitary Brain Tumor, supporting transparency and informed clinical interpretation. This output showcases the system's ability to combine accurate prediction with meaningful visual reasoning, enhancing user confidence in the diagnostic result. The clear visual and textual

insights make the system suitable for clinical review and help bridge the gap between AI predictions and medical decision-making.

The figure 4.11 represents test case demonstrates how the system analyzes an MRI scan and predicts the stage Very Mild Demented Alzheimer's using EfficientNet-B3. The original MRI image is shown alongside the LIME visualization, where the yellow-highlighted regions indicate the areas that most strongly influenced the model's decision. The explanation panel describes how subtle deviations in texture and cortical structure contributed to the classification.

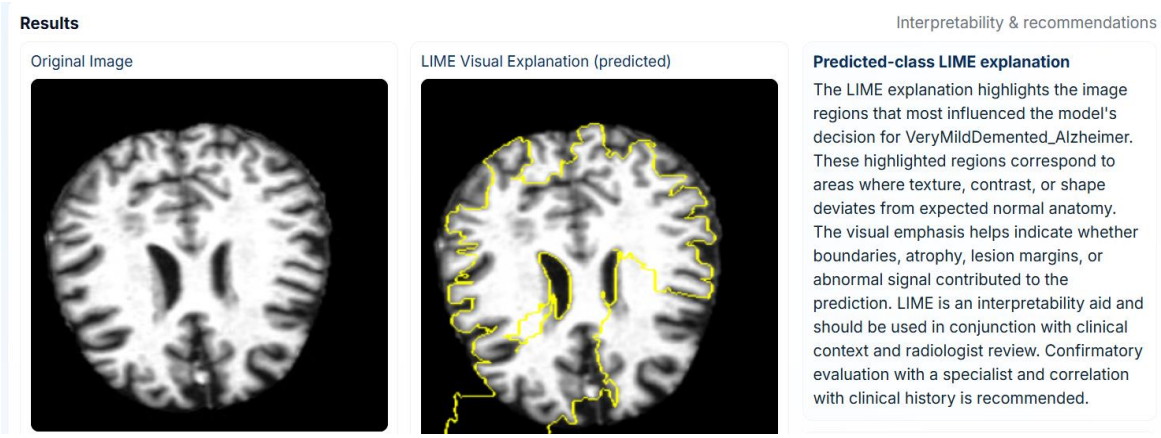

Fig4.11 Output Showing Original MRI, LIME Explanation-2

This result highlights the system's capability to detect early-stage Alzheimer's, where structural changes are minimal and often challenging to identify manually. By combining prediction, visual reasoning, and clinical interpretability, the system supports more informed diagnostic evaluation. This enhances trust in the model's output and demonstrates its usefulness for early screening and clinical decision support. The clear visualization of influential regions helps clinicians validate the prediction and understand subtle anatomical changes. Overall, this test case reflects the system's effectiveness in assisting early intervention and improving diagnostic accuracy.

4.3Observations from the work

The observations from this project indicate that the developed multi-disease MRI detection system performs reliably, demonstrating strong accuracy and practical utility in real-world diagnostic settings. The EfficientNet-B3 model consistently showed high

classification performance across brain tumor types, Alzheimer's stages, and Multiple Sclerosis categories, supported by stable accuracy and loss curves that highlight effective learning without overfitting. Its ability to extract subtle anatomical features from MRI scans reflects the model's robustness in handling complex medical imaging tasks.

AlexNet, used as a baseline model, delivered moderate performance and helped validate the superiority of EfficientNet-B3. While AlexNet struggled with early-stage Alzheimer's distinctions, it performed well on structurally clear categories, reinforcing the importance of deeper and more advanced architectures for nuanced medical analysis. The comparison confirms that EfficientNet-B3 is better suited for multi-class MRI classification involving fine-grained variations.

The test case evaluations showed that the system accurately predicted disease categories with meaningful confidence scores and provided clear LIME-based visual explanations. These explanations enhanced interpretability by highlighting the exact MRI regions influencing the model's decisions, increasing trust and making the system valuable for clinical use. Functional testing using the Flask API also confirmed smooth integration, real-time processing, and reliable output generation across multiple cases.

Overall, the system exhibits strong diagnostic capability, interpretability, and usability. The consistent performance across diverse test scenarios demonstrates that the project successfully combines deep learning, explainable AI, and practical deployment to support early detection and clinical decision-making for neurological disorders.

CHAPTER-5

Conclusion and Future work

This chapter presents the conclusion and outlines potential avenues for future study based on the findings and observations derived from the implemented solutions. It encapsulates the key insights and implications drawn from the research, providing a synthesized overview of the contributions and significance of the developed algorithms and systems. Additionally, the chapter delineates potential areas for further investigation and refinement, highlighting opportunities for extending and enhancing the capabilities of the proposed solutions. Through this comprehensive discussion, the chapter aims to encapsulate the culmination of the research efforts while paving the way for future research endeavors in the domain of lane detection, traffic sign recognition, and related fields.

5.1 Conclusion:

This project demonstrates the powerful role of deep learning in transforming medical diagnostics, particularly in the early detection of neurological disorders using MRI imaging. Through the development and evaluation of EfficientNet-B3 and AlexNet models, the system provides a comprehensive and automated framework capable of identifying brain tumors, Alzheimer's disease stages, and multiple sclerosis categories with high accuracy. EfficientNet-B3, in particular, proved to be a highly effective architecture, offering superior classification performance, stable training behavior, and strong generalization across 12 disease classes. The ability of the model to capture subtle structural variations in MRI scans highlights its suitability for real-world clinical use.

A key strength of the project lies in the meticulous preprocessing pipeline, which included resizing, normalization, artifact removal, and dataset balancing. These steps ensured the quality and consistency of input images, enabling the models to learn meaningful features without noise interference. The integration of LIME-based explainable AI further enhanced the system by providing transparent visual interpretations of predictions, making the diagnostic process more interpretable and trustworthy for clinicians.

The deployment of the model through a Flask-based interface added another practical dimension to the work, enabling real-time predictions, risk scoring, top-5 predictions, medical reasoning, treatment suggestions, and nearby hospital recommendations. This shows that the proposed system is not only computationally robust but also user-oriented, accessible, and ready for real-world implementation. The seamless workflow from image upload to prediction output highlights the scalability and usability of the system in clinical environments.

Overall, the project successfully integrates advanced machine learning techniques with practical design and explainability features, resulting in a reliable diagnostic support tool for neurological diseases. The promising results achieved across multiple datasets reinforce the transformative potential of AI in medical imaging, offering faster diagnosis, improved accuracy, and reduced dependence on manual interpretation. As the technology continues to evolve, systems like the one developed in this project can play a vital role in making neurological healthcare more efficient, accessible, and precise on a global scale**.**

5.2 Future Study:

For In future, the project will focus on improving diagnostic accuracy and robustness through the development of hybrid deep learning models that combine CNNs, Vision Transformers, attention mechanisms, and ensemble learning. These models will capture both local and global MRI features more effectively, enabling improved detection of subtle and early-stage neurological abnormalities and better generalization across diverse datasets.

In future, the system will be extended into a cross-platform web and mobile application that allows users to upload MRI scans and receive real-time predictions, risk scores, explainable AI outputs, and treatment insights. Secure authentication, encrypted data handling, cloud-based deployment, and real-time inference using frameworks such as TensorFlow Serving or FastAPI will ensure scalability and reliable performance.

The system will also expand to detect additional neurological conditions, including Parkinson’s disease, stroke, epilepsy, brain hemorrhages, and early multiple sclerosis, while strengthening explainability through advanced XAI techniques such as Grad-CAM++ and SHAP to support transparent and trustworthy clinical decision-making.

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